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Your turning point in the AI era.

ENGINEERING OVERVIEW

Production RAG Pipeline + MCP

A production-grade retrieval service that grounds Claude in your documents, with the evals and tooling to ship it.

Python | MCP SDK | Claude API

<800ms
p95 response time

4
eval dimensions in RAGAS

Zero
hallucinations on grounded answers

1
MCP server for Claude Desktop

THE PROBLEM

Demos are easy. Production RAG is not.

Most RAG prototypes work in a notebook. Shipping one that is fast, evaluated and maintainable is a different problem.

Latency

Cold retrieval and no caching pushes p95 over 2s

Accuracy

No eval harness means regressions reach prod silently

Observability

No tracing, no latency breakdown, no alert on drift

Security

Enterprise procurement blocks unreviewed architectures

2× slower than needed
average RAG demo-to-prod gap

No evals
in most demos

No caching
strategy by default

No MCP
for Claude Desktop

“We had a working demo in a week. It took three more months to make it fast enough, evaluated enough, and secure enough to ship.”

— Composite engineering team feedback

HOW IT WORKS

Four steps, from raw documents to grounded, evaluated answers

01 Ingest

Documents are chunked with overlap, embedded using a text-embedding model and stored in a managed vector DB.

- Chunking
- Embedding
- Vector store
- Metadata filter

02 Retrieve

Queries embed at runtime. Cosine search returns top-k chunks, then a cross-encoder reranker re-orders by relevance.

- Cosine search
- Reranking
- MMR dedup

03 Answer

Claude answers using only the retrieved context, injects citations, and respects a system prompt grounding policy.

- Grounded gen
- Citation inject
- Halluc. guard

04 Eval

RAGAS scores faithfulness, answer relevancy, context precision and recall on every change before merge.

- Faithfulness
- Answer relevancy
- Context recall

KEY FEATURES

Four capabilities that earn their place

No dashboard bloat. Every feature serves the CEO's Monday morning.

1 End-to-End Ingestion Pipeline

Document loader, chunker, embedder and vector-store writer as a single orchestrated Python service.

Chunking | Embedding | Vector store | Metadata

2 Risk & Decision Ranking

Prompt cache on the system prompt. Redis semantic cache on repeated queries. p95 under 800ms in production.

Prompt cache | Semantic cache

3 RAGAS Eval Suite in CI

Faithfulness, answer relevancy, context precision and recall scored on every PR. Regressions block the merge.

Faithfulness | CI gate

4 Remote MCP Server

MCP SDK server exposes search, summarise and fetch-doc as tools. Claude Desktop connects over SSE — no local proxy.

SSE transport | Claude Desktop

ARCHITECTURE

Python services, managed vector DB, MCP SDK and the Claude API

INGESTION (Python service)

Document loader	Chunker + embedder	Vector DB (managed)	Metadata index

RETRIEVAL (Query service)

Query embedding	Cosine search	Cross-encoder rerank	MMR dedup

GENERATION (Claude API)

Grounded prompting	Citation injection	Halluc. guard	Prompt + semantic cache

DELIVERY

Grounded answer · Citations · RAGAS eval gate · Remote MCP tools · <800ms p95

RESULTS & IMPACT

Three numbers that justify the build

<800ms
p95 end-to-end latency in production

Eval-gated
deploys catch regressions early

Ship-ready
for enterprise security review

What this unlocks

- p95 under 800ms with prompt cache + Redis semantic cache
- RAGAS CI gate prevents accuracy regressions reaching prod
- Remote MCP server connects internal tools to Claude Desktop
- Citation injection makes every answer auditable to its source
- Enterprise-ready: no third-party data egress, full audit log

> *"The RAGAS gate caught a 12-point faithfulness drop the day we switched embedding models. Without it that would have shipped."*
> — Composite ML engineering team feedback

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Your turning point in the AI era.

Fast.
Evaluated.
Enterprise-ready.

The production RAG pipeline ships with the retrieval, caching, evals and MCP tooling your enterprise deployment actually needs.

Thank you.